

Mid Semester Test-III, November 2022

BE III Semester, Information Technology (Section: A and B)

ITR3G1: Computer Organization and Architecture

Max. Marks: 20

Total No. of Questions: 04

Note: All questions are compulsory. Make suitable assumptions if necessary. The answer should be to the point.

Q.1 Describe the following terminologies associated with pipeline computers: 05

- (i) Static Pipeline (ii) Minimum Average Latency (iii) Multifunctional Pipeline
(iv) Forbidden Latencies (v) Collision Vector

Q.2 Derive the speedup of a **k-stage** linear pipeline processor over an equivalent nonpipeline processor. 05

Q.3 Draw the interconnection structures and data flow patterns in general pipelines with either feedforward or feedback connections, in addition to the cascaded connections in a linear pipeline. Assume that the pipeline is dual functional, denoted as function A and function B. Also number the pipeline stages S1, S2, and S3 from the input end to the output end. Use the reservation table to show the utilization of successive pipeline stages. 05

Q.4 Classify the pipeline processors according to the: 05
(iii) Handler
(iv) Ramamoorthy and Li